# Kaicheng Xue

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## RESEARCH INTERESTS

Understanding and evaluating the behavior of large language models, from their internal mechanisms to their use as agents in the real world.

## EDUCATION

**University of Virginia, Charlottesville, VA** Aug 2025 – Dec 2026 (expected)

M.S. in Computer Science, GPA: 4.0/4.0

*Evaluating Entropy and Spectral Gap of Hidden-State Markov Chains as Predictors of LLM Reasoning Accuracy*

Thesis advisor: Yangfeng Ji

Relevant Coursework: Machine Learning, Natural Language Processing, Trustworthy AI, Probabilistic Machine Learning, Data Privacy, Neural Networks, Geometry of Data

**University of Wisconsin–Madison, Madison, WI** Sept 2021 – May 2025

B.S. in Mathematics and Computer Science (double major), GPA: 3.79/4.0 (Mathematics 3.81, CS 3.78)

Relevant Coursework: Introduction to Algorithms, Introduction to Artificial Intelligence, Real Analysis, Stochastic Processes, Probability, Abstract Algebra, Linear Optimization, Linear Algebra, Discrete Mathematics

Honors: College of Letters & Science Dean's List

## PUBLICATIONS

**Xue, K.**, Zhang, X., Mao, R., & Cambria, E. "Understanding the Hidden State of Language Models from the Perspective of Sentiment Analysis." In *Proceedings of the 2024 IEEE International Conference on Data Mining Workshops (ICDMW)*, 2024. *(Workshop paper)*

## RESEARCH EXPERIENCE

**Graduate Research, University of Virginia** Jan 2026 – Nov 2026

*Evaluating Entropy and Spectral Gap of Hidden-State Markov Chains as Predictors of LLM Reasoning Accuracy*

Advisor: Yangfeng Ji

- Modeled hidden-state trajectories of nine Llama and Qwen3 models over reasoning steps as Markov chains, using entropy rate and spectral gap as accuracy predictors
- Evaluated the predictors on four reasoning benchmarks with nested cross-validation and permutation tests, including LoRA-fine-tuned variants
- Found that model size and family explain most accuracy variation on two of four datasets, and that the discovered states mostly track a step's position

**Student Researcher, University of Virginia** Aug 2026 – Present

*LLM Abuse in Digital Advertising Marketplaces: Detection and Mitigation*

PI: Tianhao Wang (CS); Co-PIs: Michael Albert (Darden), Jingjing Li (McIntire)

- Build a simulated Amazon-style marketplace populated by LLM shopper agents, with personas matched to real Amazon buyers
- Implement an LLM-generated fake-review attack that combines push, segment-targeted, and average-profile reviews, and compare shopper behavior with and without it
- Building on real shopper data, explore how to make shopper agents behave more like humans

**Course Research Project, University of Virginia** Spring 2026

*Privacy Attack via Temporal Embedding Inversion*

Course: Data Privacy; Instructor: Tianhao Wang

- Proposed a sliding-window method that uses dialogue history to invert conversation embeddings leaked from RAG systems
- Raised attack success over the baseline, e.g., from 15% to 41% on user profiles, on a dataset of profile, financial, and medical conversations
- Analyzed why a non-linear extension fails to beat the linear method, attributing it to distorted embedding-space geometry

**Research Intern, Nanyang Technological University** May 2024 – Sept 2024

*Understanding the Hidden State of Language Models from the Perspective of Sentiment Analysis*

Supervisors: Xulang Zhang, Rui Mao, Erik Cambria

- Analyzed hidden states of a fine-tuned RoBERTa across layers and attention heads with K-means, spectral, and hierarchical clustering
- Found that clustering separates sentiment mainly in the last three layers (about 80% accuracy at layer 12 versus about 50% earlier)
- Showed that early-to-mid layers encode part-of-speech information, and that across word sense disambiguation and entailment the mechanisms are task-driven rather than human-like

## TECHNICAL SKILLS

**Languages:** Python, Java, C, SQL, JavaScript
**Frameworks & Tools:** PyTorch, Hugging Face Transformers, scikit-learn, NumPy, pandas, MATLAB
**Infrastructure:** Git, Linux, CUDA, Docker, AWS, LaTeX
**Methods:** Hidden-state probing, Markov chain analysis, stochastic processes, spectral analysis, clustering (K-means, spectral, hierarchical), benchmark construction, statistical evaluation (permutation tests, cross-validation)